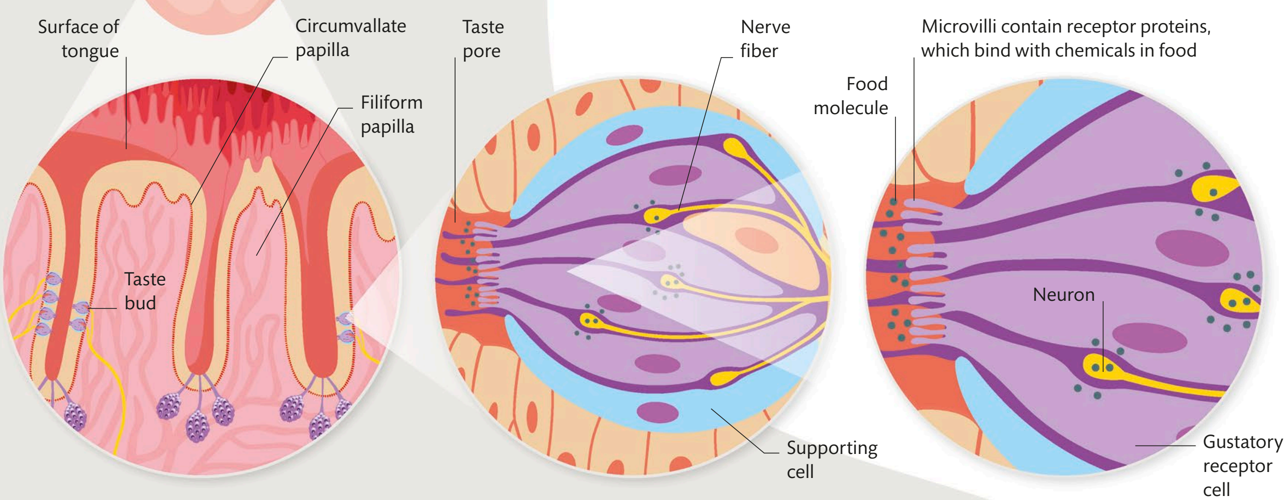
Taste

Fueling the body requires the intake of nourishing foods and liquids. Choosing what is safe to eat is largely influenced by our senses of taste and smell.

Picking up taste

Taste is actually a limited sense; there are only five basic tastes that can be detected (see right). Like smell, taste is a chemosense. Chemical substances in food are picked up by the taste buds, which are mainly found on the tongue. Receptor cells, housed in structures called microvilli within the taste bud, detect these chemicals and send signals to the brain for processing.

1 Tongue
The tongue is a strong, flexible muscle. It is used to push food around the mouth and for speech. Its upper surface is covered in tiny projections called papillae. Most of the papillae are filiform, or threadlike, structures and contain no taste buds. They help grip and wear down food while it is being chewed.



2 Papillae
In addition to filiform papillae, the tongue has fungiform (mushroomlike), foliate (leaflike), and circumvallate (wall-like) papillae, which all contain taste buds. Most taste buds are found in the foliate papillae on the back and sides of the tongue.

3 Taste buds
A taste bud is a collection of 50–100 cells that are clustered together like segments in an orange. They are located in the walls of papillae. One end of each cell protrudes out of the bud, where it gets washed with saliva containing food molecules.

4 Taste bud cells
When food molecules hit the cells, they interact with either receptor proteins or porelike proteins called ion channels. This causes electrical changes in the cell, which prompt neurons at the base of the cell to send signals to the brain.

The five basic tastes

Taste is an evolutionary adaptation for survival. Being able to determine whether something is nutritious or potentially poisonous before taking it into the body is enormously important. So far, only five basic tastes have been discovered, although there may be others.

Sweet
Signals the presence of carbohydrates, which are sources of vital sugars.

Salty
Detects chemical salts and minerals that are needed by the body.

Sour
Warns against foods that may be unripe or going bad.

Bitter
Poisons and other toxins are often bitter or unpalatable.

Umami
Detects glutamate salts and amino acids, which are found in meat, cheese, and other aged or fermented foods.